11.2 Fundamental Particles

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.2 Fundamental Particles
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What is a particle that is **not** made of any smaller particles?

A. elementary particle

B. minimum particle

C. original particle

D. atomic particle

[1 mark]

Question 2

Light is made up of which type of elementary particle?

A. quarks

B. leptons

C. photons

D. bosons

[1 mark]

Question 3

What type of elementary particles are electrons ?

A. quarks

B. leptons

C. photons

D. bosons

[1 mark]

Question 4

What elementary particles are the building blocks for protons and neutrons ?

A. quarks

B. leptons

C. photons

D. bosons

[1 mark]

Question 5

How many quarks make up a proton ?

A. 0

B. 1

C. 2

D. 3

[1 mark]

Question 6

Which particle is made up of two 'down' quarks and one 'up' quark?

A. atom

B. electron

C. neutron

D. proton

[1 mark]

Question 7

How many types of quark are there?

A. 2

B. 4

C. 6

D. 8

[1 mark]

Question 8

Which elementary particle is a lepton?

A. neutron

B. electron

C. quark

D. proton

[1 mark]

Question 9

What are the fundamental particles of an atom?

A. quarks, gluons and electrons

B. protons, neutrons and electrons

C. quarks, photons and electrons

D. quarks, gluons and neutrino

[1 mark]

Question 10

What are mesons ?

A. a type of composite particle produced by high energy

B. a type of gluon

C. an antimatter version of the electron

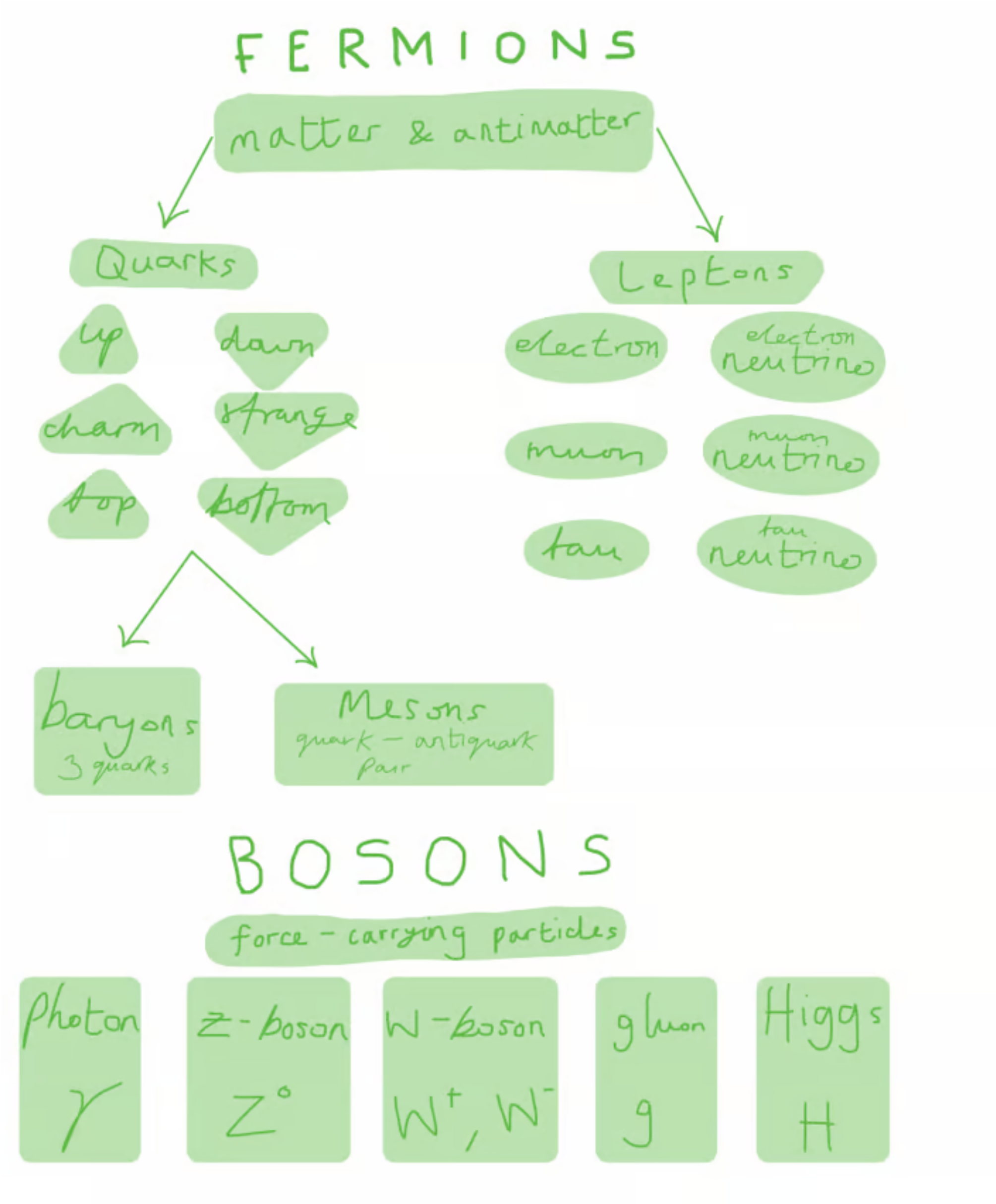
D. a type of neutrino

[1 mark]

Model Answers: Easy

1

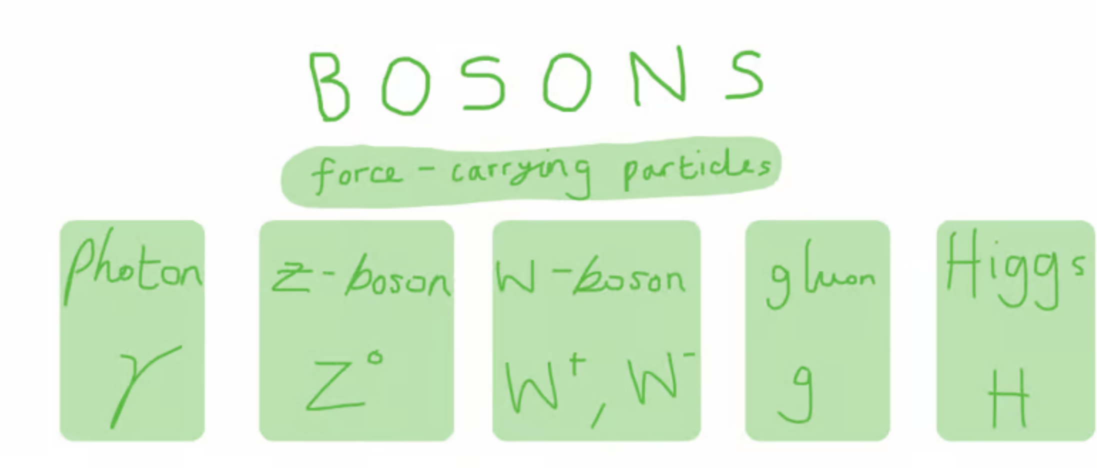
The correct answer is **A** because an **elementary particle** or **fundamental particle** is a subatomic particle which cannot be broken down further into constituent particles



2

The correct answer is **C** because:

- Photons are the smallest discrete amount or quantum of electromagnetic radiation
  - It is the basic unit of all light
- Photons are always in motion and travel at 3.0×10^8 m s^{-1} in a vacuum
- Photons are a type of boson



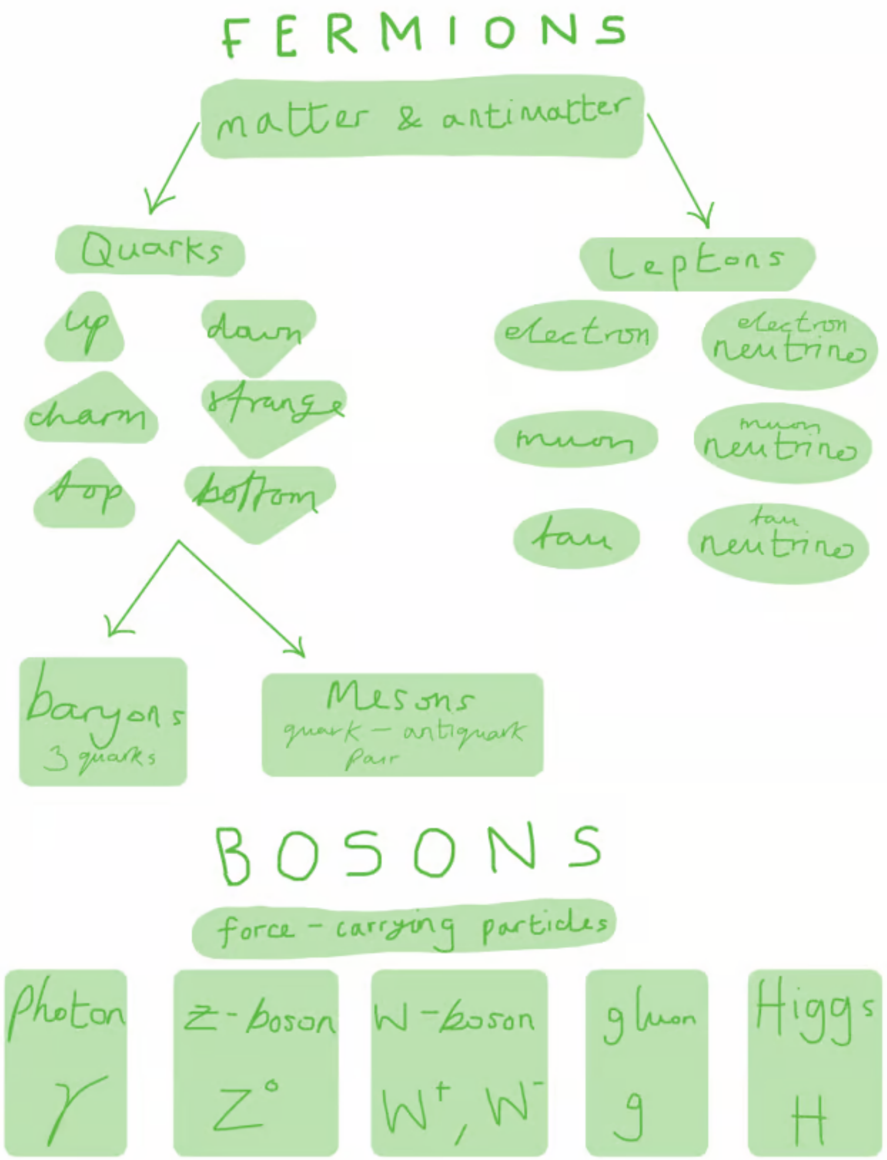
A is incorrect because quarks are fermions which cannot exist on their own, instead they combine to make hadrons (e.g. protons, neutrons)

B is incorrect because leptons are fermions (e.g. electrons, neutrinos)

D is incorrect because bosons are force-carrying particles (e.g. photons, gluons)

3

The correct answer is **B** because electrons are a type of **lepton**



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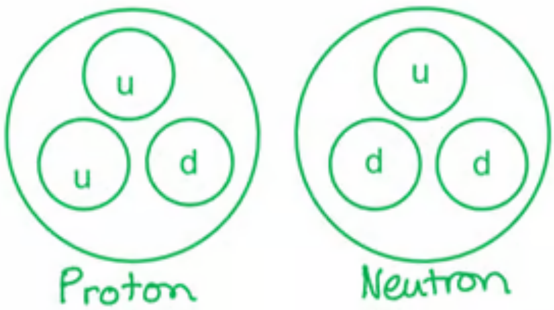
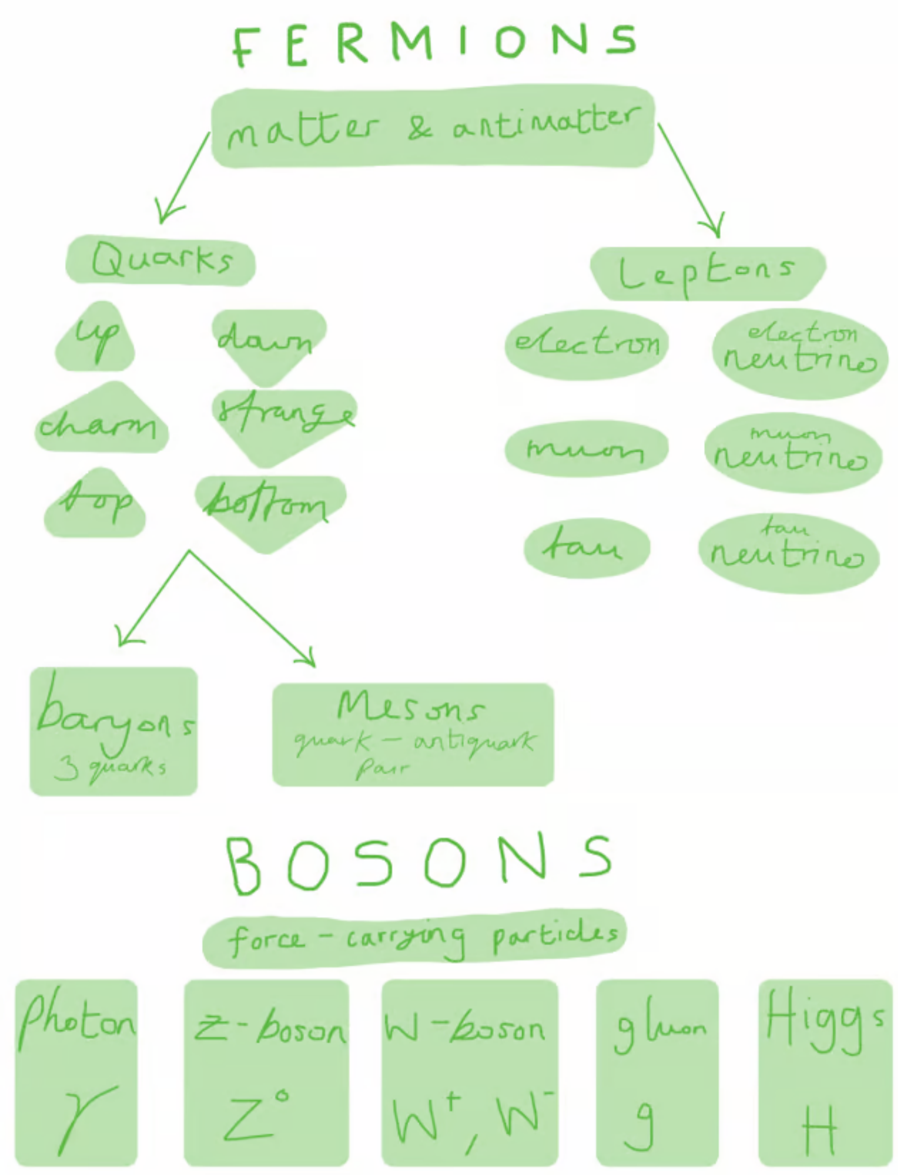
The correct answer is **A** because:

- Quarks are the building blocks for protons and neutrons
- Protons and neutrons are composed of two types of quarks (up and down) held together by gluons :

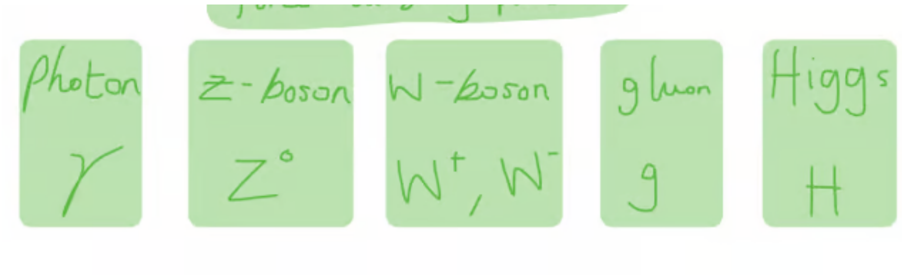




5

The correct answer is **D** because:

- Protons and neutrons are composed of two types of quarks held together by gluons:
    - Up quarks
    - Down quarks
- They are both made of three quarks as shown in the diagram



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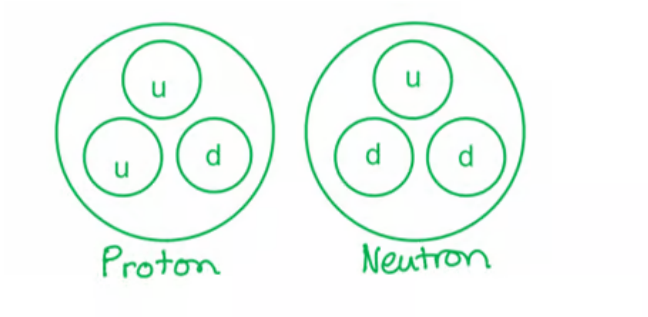

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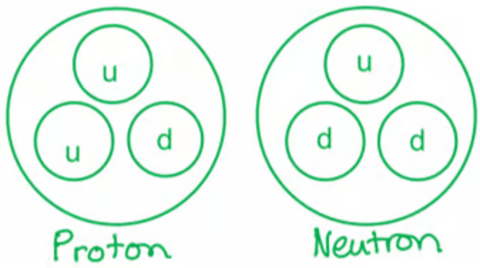
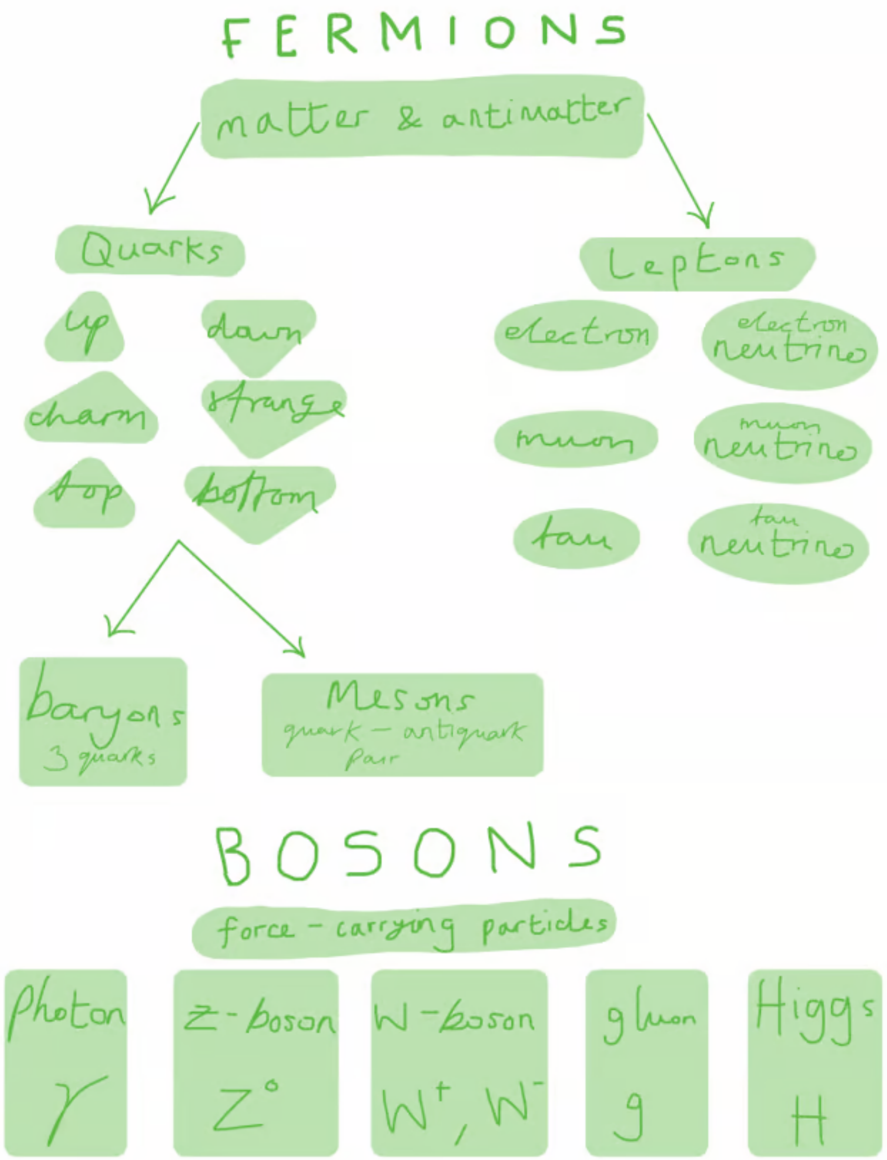


- Therefore, neutrons consist of two 'down' quarks and one 'up' quark

7

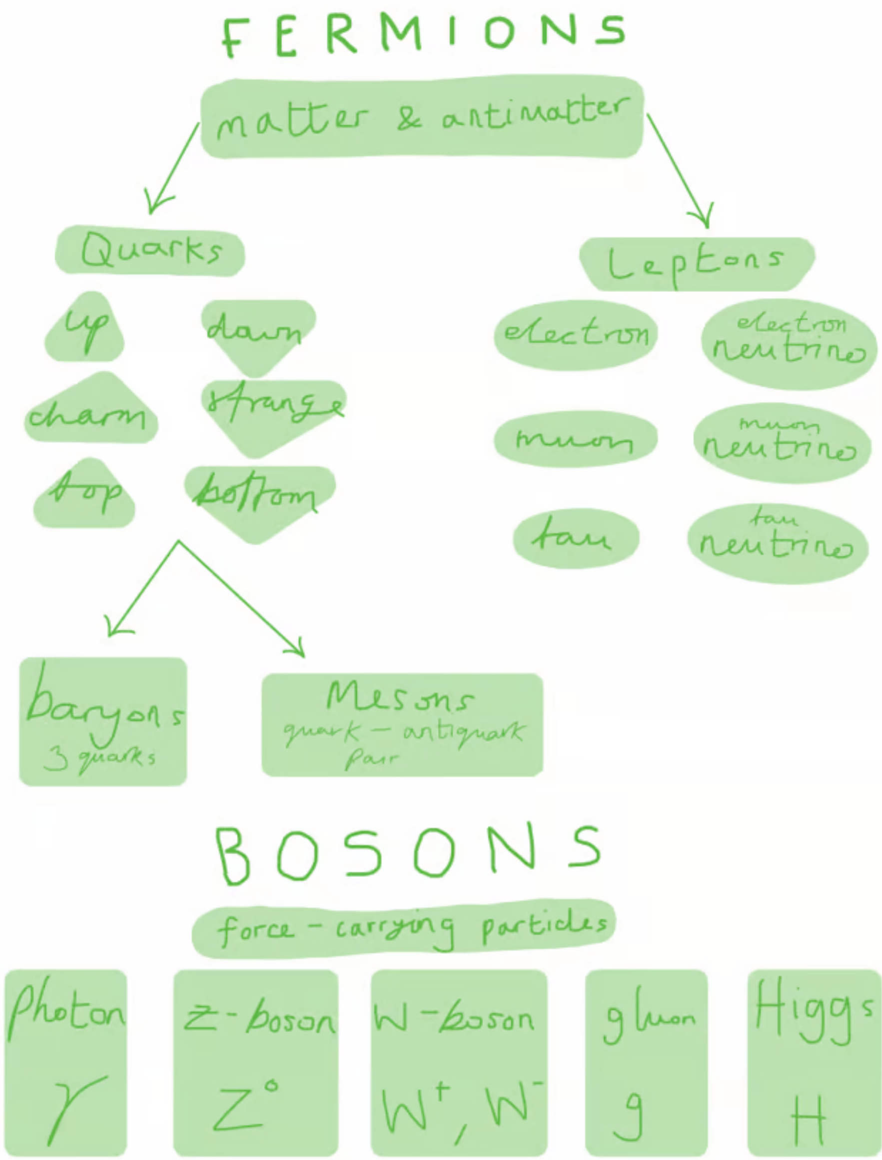
The correct answer is **C** because:

- There are six types, or flavours , of quarks that differ from one another in their mass and charge characteristics
- These **six quark** flavours can be grouped in three pairs:
  - Up and down, charm and strange, and top and bottom



8

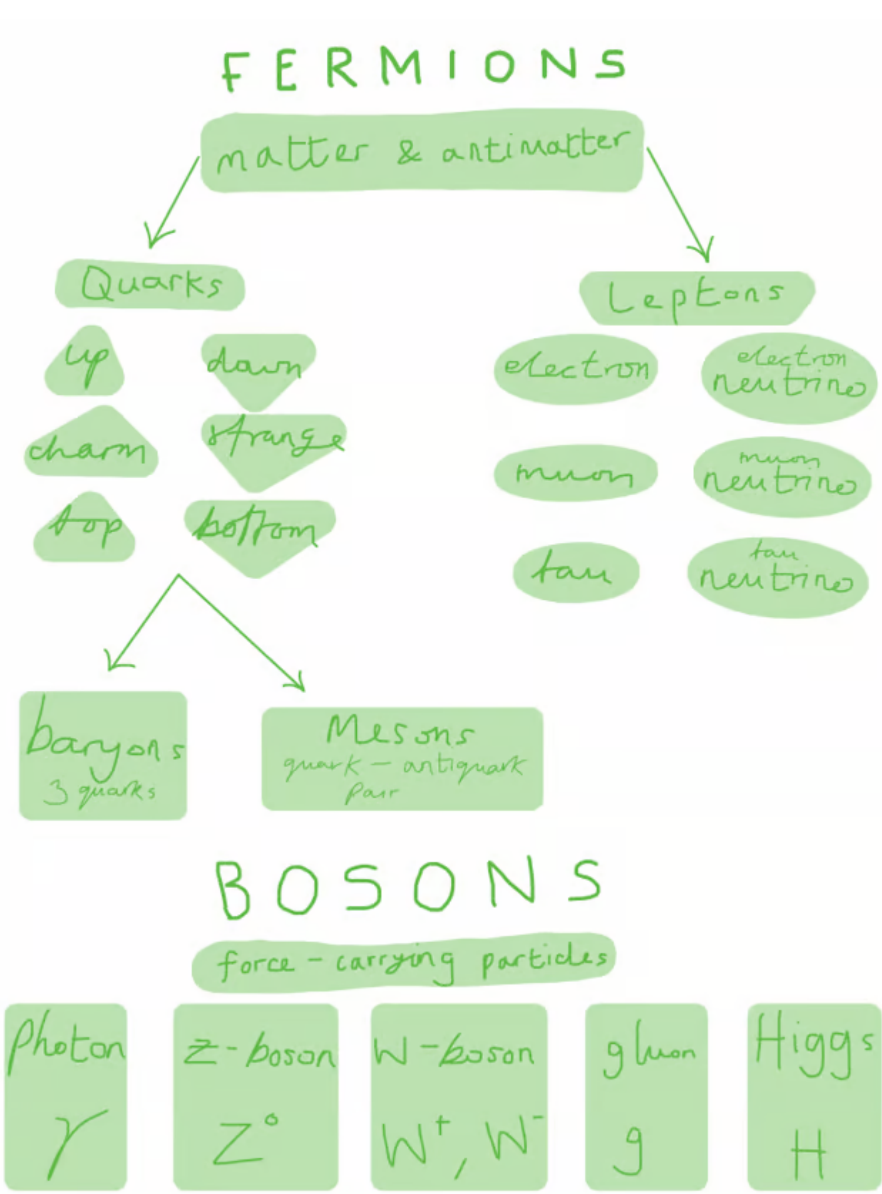
The correct answer is **B** because electrons are a type of **lepton**



9

The correct answer is **A** because:

- Atoms are made up of **elementary particles** or **fundamental particles**
- The elementary particles that make up an atom are quarks, gluons, and electrons
  - Quarks and gluons that make up protons and neutrons and are found in the nucleus of the atom
  - Gluons are a type of boson (force-carrying particle)
  - Electrons are a type of lepton which orbit around the nucleus
- The 12 fundamental particles are as follows :



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- The 12 fundamental particles are as follows :

B is incorrect because protons, neutrons and electrons are not all elementary particles

C is incorrect because photons are not a fundamental particle in an atom

D is incorrect because neutrinos are not a fundamental particle in an atom

10

The correct answer is **A** because:

- Hadrons are subatomic particles made from 2 or more quarks which are bound by gluons
- There are two types of hadron:
  - Baryons: three quarks or three antiquarks
  - Mesons: a quark and an anti-quark pair
- They only exist for short periods of time and are produced through very high energy interactions such as those caused by cosmic rays or in particle accelerators
- Therefore, mesons are a type of composite particle produced by high energy

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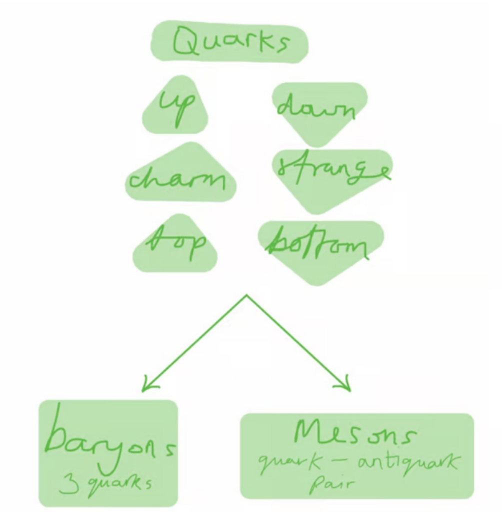

B is incorrect because gluons are a type of boson (force-carrying particle)

C & D are incorrect because neutrinos and anti-electrons (positrons) are types of lepton